

A Complete Signed Two-Peakon Scattering and Collision Atlas for the Camassa–Holm Equation

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Abstract

On the ordered two-peakon manifold of the Camassa–Holm equation, momentum and energy reduce the four-dimensional flow to one separable quadratic for the exponential gap. The two root orders give explicit global same-sign scattering and a finite peakon–antipeakon collision with a quadratic gap and reciprocal amplitude blow-up.

1 Frozen equation and reduction

We use the momentum form

$$m_t + um_x + 2u_xm = 0, \quad m = u - u_{xx}, \quad (1)$$

and the ordered ansatz

$$u(x, t) = \sum_{j=1}^2 p_j(t) e^{-|x - q_j(t)|}, \quad q_1 < q_2. \quad (2)$$

This is the normalization introduced with peaked solutions in Camassa and Holm [1]. Since $(1 - \partial_x^2)e^{-|x - q|} = 2\delta_q$, matching the δ'_q and δ_q coefficients in (1) gives

$$\dot{q}_i = \sum_j p_j e^{-|q_i - q_j|}, \quad \dot{p}_i = p_i \sum_j p_j \operatorname{sgn}(q_i - q_j) e^{-|q_i - q_j|}. \quad (3)$$

Put $q = q_2 - q_1$, $y = e^q$, $p = p_2 - p_1$, and

$$P = p_1 + p_2, \quad E = p_1^2 + p_2^2 + 2p_1p_2e^{-q}, \quad D^2 = 2E - P^2. \quad (4)$$

Lemma 1 (Invariant scalar reduction). *Before collision, P and E are constant and*

$$\dot{y} = p(y - 1), \quad \dot{p} = \frac{P^2 - p^2}{2y}, \quad \dot{y}^2 = D^2(y - 1) \left(y - \frac{P^2}{D^2} \right). \quad (5)$$

Proof. In the ordered chart, (3) says $\dot{p}_1 = -p_1p_2e^{-q}$ and $\dot{p}_2 = p_1p_2e^{-q}$. Direct differentiation proves $\dot{P} = \dot{E} = 0$. Substitution of $p_1 = (P - p)/2$, $p_2 = (P + p)/2$ into (4) gives

$$2E = P^2 + p^2 + \frac{P^2 - p^2}{y},$$

or $p^2(y - 1) = D^2y - P^2$. Combining this with $\dot{y} = p(y - 1)$ gives the last identity in (5); differentiating the algebraic identity gives the middle one. $\square$

Theorem 1 (Complete ordered two-peakon atlas). *For $q > 0$,*

$$P^2 - D^2 = 4p_1p_2(1 - e^{-q}).$$

Consequently, every ordered solution with $p_1p_2 \neq 0$ belongs to exactly one of the following strict chambers. The equality face $p_1p_2 = 0$ is the degenerate single-peak boundary (including the zero field), not a third strict two-body chamber.

Same sign: if $P^2 > D^2$, there is a unique time shift t_ such that*

$$y = 1 + \left(\frac{P^2}{D^2} - 1 \right) \cosh^2 \frac{D(t - t_*)}{2}, \quad p = D \tanh \frac{D(t - t_*)}{2}. \quad (6)$$

The solution is global and the gap remains positive.

Opposite sign: if $D^2 > P^2$, an incoming branch has a finite collision time t_c and

$$y = 1 + \left(1 - \frac{P^2}{D^2}\right) \sinh^2 \frac{D(t_c - t)}{2}, \quad p = -D \coth \frac{D(t_c - t)}{2}. \quad (7)$$

As $t \uparrow t_c$,

$$q = \frac{D^2 - P^2}{4} (t_c - t)^2 + O((t_c - t)^4), \quad p = -\frac{2}{t_c - t} + O(t_c - t). \quad (8)$$

Proof. The second root of the quadratic in (5) is P^2/D^2 . When it exceeds one, separation with the minimum at t_* gives (6). When it lies below one, separation from $y = 1$ gives (7). Differentiation verifies both branches. The expansions $\sinh z = z + O(z^3)$ and $\coth z = z^{-1} + O(z)$ prove (8). $\square$

2 Scattering, collision, and the profile limit

For the same-sign branch, one may choose the additive centre constant so that

$$q_1 + q_2 = Pt + 2 \operatorname{sgn}(P) \operatorname{artanh} \left(\frac{D}{|P|} \tanh \frac{D(t - t_*)}{2} \right). \quad (9)$$

Thus the asymptotic momenta are $(P - D)/2$ and $(P + D)/2$, and the labeled amplitudes exchange these values. This is scattering, not a periodic-orbit family.

In the signed chamber, put $c = (q_1 + q_2)/2$ and $h = q/2$. Then

$$\dot{c} = \frac{P}{2} (1 + e^{-q}),$$

so finite collision time implies $c \rightarrow q_c$. For $K_a(x) = e^{-|x-a|}$, the kernel is uniformly one-Lipschitz in its centre and

$$u = \frac{P}{2} (K_{c-h} + K_{c+h}) + \frac{p}{2} (K_{c+h} - K_{c-h}), \quad \|u - PK_c\|_\infty \leq (|P| + |p|)h.$$

By (8), $h = O((t_c - t)^2)$ and $p = O((t_c - t)^{-1})$; hence the last bound tends to zero. Together with $c \rightarrow q_c$, this proves uniform convergence to $Pe^{-|x-q_c|}$. The limiting ordinary single-peak energy is P^2 , whereas the pre-collision invariant is

$$E = \frac{P^2 + D^2}{2};$$

the excess $(D^2 - P^2)/2$ is the concentrated collision-energy ledger. Explicit non-symmetric dissipative peakon-antipeakon continuations are studied by Grunert and Holden [2]; we next state precisely the continuation used here.

References

- [1] R. Camassa and D. D. Holm, *An integrable shallow water equation with peaked solitons*, Physical Review Letters **71**(11) (1993), 1661–1664, [doi:10.1103/PhysRevLett.71.1661](https://doi.org/10.1103/PhysRevLett.71.1661).
- [2] K. Grunert and H. Holden, *The general peakon-antipeakon solution for the Camassa–Holm equation*, Journal of Hyperbolic Differential Equations **13** (2016), 353–380, [doi:10.1142/S0219891616500119](https://doi.org/10.1142/S0219891616500119).